

RESEARCH ARTICLE SUMMARY

SINGLE-CELL ANALYSIS

Stereo-cell: Spatial enhanced-resolution single-cell sequencing with high-density DNA nanoball-patterned arrays

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INTRODUCTION: Single-cell sequencing has transformed our ability to study cellular diversity and molecular function. However, most existing methods require trade-offs between throughput, sensitivity, and compatibility with diverse cell types. Limitations in capture uniformity and compatibility with different cell sizes or experimental modalities restrict their utility in applications such as unbiased cell detection, analysis of in situ microenvironments or cell-cell interactions, and integration with spatial or multimodal assays.

RATIONALE: We hypothesized that high-density DNA nanoball (DNB)-patterned arrays, with their planar architecture and nanometer-scale resolution, could enable a single-cell sequencing strategy based on in situ transcript capture without the need for compartmentalized encapsulation. This design would support direct profiling of small to large cells, extracellular vesicles, and microstructures. To this end, we developed Stereo-cell, a spatially resolved, high-throughput single-cell transcriptomic platform built on DNB arrays. This system enables scalable, unbiased capture of cells across a wide input range and supports high-fidelity transcriptome profiling while also allowing integration with multiplexed imaging and multiomics workflows.

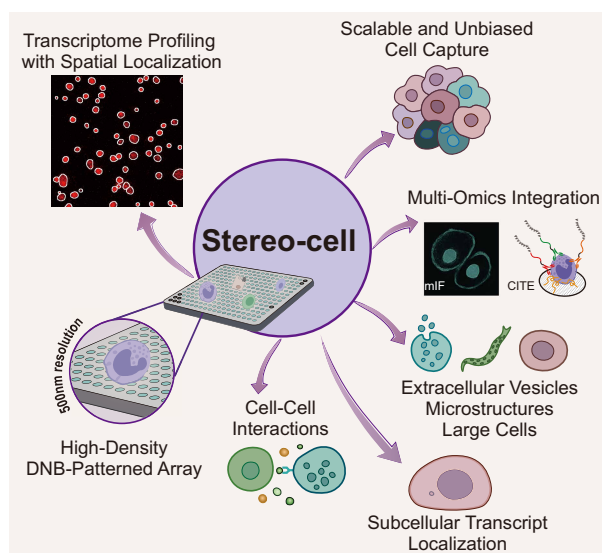
RESULTS: Stereo-cell accurately and efficiently captured transcriptomes from a broad range of cell types, with input sizes spanning from as few as 200 to nearly 1 million cells per chip. The platform demonstrated high reproducibility and robust performance across different cell loading densities, with consistent transcript detection and minimal doublet rates, supported by deep learning-based cell segmentation and imaging-based doublet elimination. In benchmarking studies that used human peripheral blood mononuclear cells (PBMCs), Stereo-cell generated gene

expression profiles closely matching those obtained from established droplet-based methods while recovering cell-type proportions that aligned more faithfully with flow cytometry reference data. Notably, Stereo-cell identified rare immune populations, such as hematopoietic stem and progenitor cells, from large-scale samples. Integration with multiplex immunofluorescence and oligo-barcoded antibodies enabled concurrent profiling of mRNA and protein markers, revealing distinct immune activation states and uncovering phenotypic diversity not apparent from RNA alone. Stereo-cell further enabled in situ profiling of cultured fibroblasts, capturing microenvironmental cues, extracellular vesicle-like structures based on distinct gene signatures and imaging data, and fibroblast responses to stimulation. Stereo-cell succeeded in sequencing structurally complex and oversized samples, including mouse oocytes and multinucleated skeletal myofibers, while preserving transcript localization at subcellular resolution. RNA staining validation and spatial gene module analysis confirmed Stereo-cell's capability to resolve intracellular transcript architecture and microenvironmental organization.

CONCLUSION: Stereo-cell establishes an integrative framework for high-throughput single-cell transcriptomics with spatial and multimodal capabilities. This platform overcomes key limitations of existing methods, improving scalability, flexibility, and precision in profiling cell-cell interactions, microenvironments, and subcellular gene expression. These improvements underscore the expansive potential of Stereo-cell across a spectrum of applications. □

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Stereo-cell: A single-cell multimodal platform. Stereo-cell leverages high-density DNB-patterned arrays for scalable and unbiased cell capture, enabling single-cell transcriptome profiling, integration with multiomics workflows as well as in situ analysis of cell-cell interactions, structurally complex and oversized samples, and subcellular transcript localization. mIF, multiplex immunofluorescence. CITE, cellular indexing of transcriptomes and epitopes.



SINGLE-CELL ANALYSIS

Stereo-cell: Spatial enhanced-resolution single-cell sequencing with high-density DNA nanoball-patterned arrays

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Single-cell sequencing technologies have advanced our understanding of cellular heterogeneity and biological complexity. However, existing methods face limitations in throughput, capture uniformity, cell size flexibility, and technical extensibility. We present Stereo-cell, a spatial enhanced-resolution single-cell sequencing platform based on high-density DNA nanoball (DNB)-patterned arrays, which enables scalable and unbiased cell capture at a wide input range and supports high-fidelity transcriptome profiling. Stereo-cell further allows integration with imaging-based modalities and multiomics strategies, including immunofluorescence and epitope profiling. This platform is also compatible with profiling extracellular vesicles, microstructures, and large cells, whereas its spatial resolution facilitates *in situ* analysis of cell-cell interactions, cellular microenvironments, and subcellular transcript localization. Together, Stereo-cell provides a flexible framework for expanding single-cell research applications.

In the past decade, single-cell sequencing has transformed our understanding of cellular heterogeneity and functional diversity, enabled by technologies that profile the genome, epigenome, and transcriptome at single-cell resolution (1–4). Although powerful, most single-cell approaches face a number of technological constraints. For example, although plate-based single-cell technologies, such as SMART-seq (switching mechanism at the 5' end of the RNA template), have high

detection sensitivity, the cell throughput is limited (5, 6). The single-cell combinatorial indexing strategy offers high throughput, but the sensitivity is sacrificed in addition to the laborious experimental procedure (7–9). On the other hand, droplet barcoding is now the most widely used strategy for single-cell RNA sequencing (scRNA-seq), exemplified by the 10× Genomics Chromium platform that enables the sequential encapsulation of thousands of cells into nanoliter-scale droplets with a microfluidic instrument (10–12). However, scaling up the throughput is challenging, largely owing to the limitation in the droplet generation speed. To address this issue, an instrument-free method, particle-templated instant partition sequencing (PIP-seq), has been developed for simultaneous encapsulation to profile a wide range of cell numbers from tens to millions (13).

As single-cell research rapidly advances toward whole-organism scale under complex conditions (14–17), there is a growing demand to further enhance the performance and scope of the application for single-cell sequencing methods. To this end, several directions of technological improvement need to be considered, including: (i) extension of the detection scope for both large-scale single-cell studies and research on rare cells, which are usually available only in small quantities; (ii) acquiring the genomic or transcriptomic profiles together with the optic signals that enable recognition of cell morphology, cell surface signals, and marker protein expression; (iii) detection of cell-cell interactions and microenvironment *in situ* under natural cell growth conditions; (iv) capability to capture cells or partially dissociated microstructures of large sizes and various shapes; and (v) simultaneous determination of the identity and subcellular distribution of each transcript for RNA sequencing.

By combining DNA nanoball (DNB)-patterned arrays and *in situ* RNA capture, we previously created spatial enhanced-resolution omics sequencing (Stereo-seq), which enables *in situ* transcriptome profiling of frozen tissue sections placed onto the chip surface (18, 19). The DNB chip features densely packed nanoballs (~220 nm in diameter, spaced 500 nm from center to center) that facilitate encapsulation-free capture of both cells and transcripts. This design achieves single-cell spatial resolution in transcript detection, directly mapping cellular molecular signatures to their native tissue context.

In this work, we develop and report the spatial enhanced-resolution single-cell sequencing (Stereo-cell) based on DNB-patterned arrays. We show that Stereo-cell generates high-quality transcriptome profiles comparable to or even better than mainstream single-cell sequencing technologies, with a wider range of input cell numbers and unbiased recovery of individual cell populations. Similar to Stereo-seq for tissue profiling, Stereo-cell is compatible with immunofluorescent staining and imaging as well as multiomics analysis, which not only allows for identification and removal of homotypic and heterotypic doublets but also assures the accurate assignment of transcripts and proteins to each cell simultaneously. More notably, Stereo-cell can identify subcellular localization for certain RNA molecules, cellular microenvironments, and multicellular communities. Stereo-cell overcomes a number of technical limitations for single-cell sequencing approaches and holds the potential for a wide range of applications.

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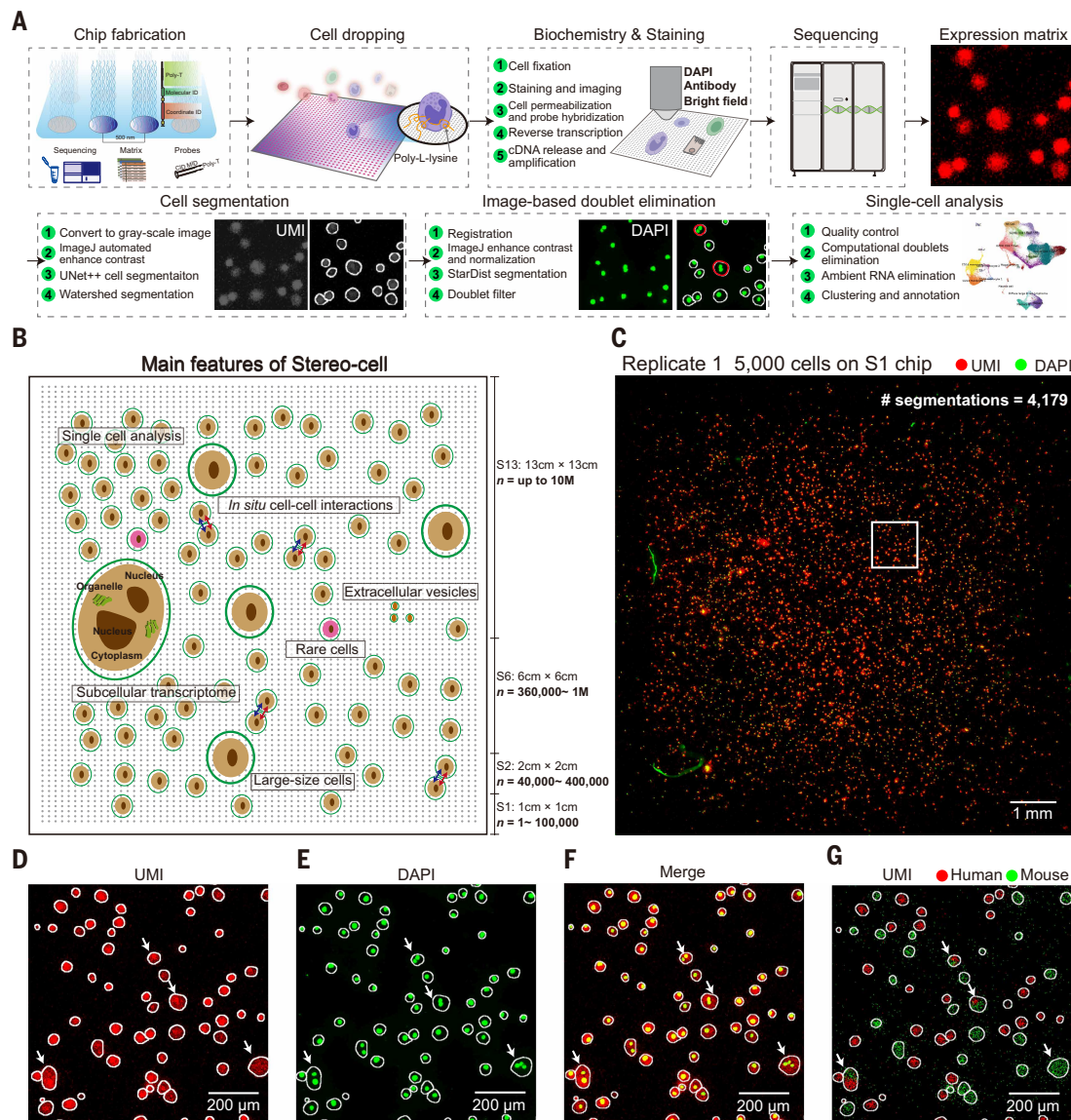


Fig. 1. An overview of pipeline and expanding capacity in Stereo-cell technology. (A) Experimental outline and analysis workflow for the Stereo-cell process. (B) Schematic showing the potential characteristics of Stereo-cell. (C) Spatial visualization of the superimposed nuclei staining (green, DAPI) and distribution of captured transcripts (red, UMIs) on an S1 chip with an input of 5000 cells at a 1:1 ratio of human (HEK293T) and mouse (NIH-3T3) cell lines. Scale bar, 1 mm. (D to G) Magnification showing the captured transcript signals (D), DAPI nuclei staining (E), superimposed DAPI and UMIs (F), and distribution of captured human (red) and mouse (green) transcripts (G) from the region squared in (C). The arrows show the single-cell segmentation where multiple nuclei were present. Scale bar, 200 μ m.

Results

High-throughput capture of single-cell transcriptomes using DNB-patterned arrays

Taking advantage of the nanometer resolution and scalable field of view for DNB-patterned arrays (DNB chips), we aimed to develop a strategy for high-throughput profiling of gene expression in single cells (Fig. 1A, chip fabrication). To this end, we first precoated the DNB chips with poly-L-lysine (PLL) to enable the attachment of suspended cells by enhancing the electrostatic interaction between the negatively charged cell membrane and the surface of the DNB chip (Fig. 1A, cell dropping). The attached cells were further processed by the standard Stereo-seq workflow (18) with a few modifications (Fig. 1A, biochemistry & staining and sequencing) (see materials and methods). After the routine data processing, the spatially resolved expression matrix was obtained (Fig. 1A, expression matrix), which was transformed and converted into a grayscale image for cell segmentation (Fig. 1A, cell

segmentation). Moreover, such cell imaging information provided by Stereo-cell ensured the accurate removal of doublet cells for the following single-cell analysis (Fig. 1A, image-based doublet elimination and single-cell analysis). Owing to its fundamentally distinct workflow from preceding single-cell sequencing technologies (such as microfluidic-based platforms) (10–12), Stereo-cell aims to overcome existing technical constraints by expanding the spectrum of cell throughput, ensuring compatibility with cell culture and imaging, and overcoming cell size restrictions to enable the investigation of in situ cell microenvironment, cell-cell interactions, and the subcellular localization of transcripts of interest (Fig. 1B). These potential features have been evaluated and presented in the following sections.

To analyze the capture efficiency and accuracy of spatial assignment for transcripts detected by Stereo-cell, we loaded a 1×1 -cm chip (S1 chip) with a 1:1 mix of 2500 human [human embryonic kidney (HEK) 293T] and 2500 mouse (NIH-3T3) cells. The resultant RNA signal

distributions visually corresponded to each individual cell and were highly colocalized with 4',6-diamidino-2-phenylindole (DAPI) signals (Fig. 1, C to F). This allowed us to separate transcripts in one RNA aggregate from neighboring capture spots and perform single-cell segmentation using a deep learning-based approach (Fig. 1D) (see materials and methods) (20). This experiment yielded 4179 (2045 human and 1860 mouse cells) segments on the chip. In general, a median of 4286 (human) and 4999 (mouse) DNB spots were captured within each cell, along with 10,265 (human) and 8896.5 (mouse) unique molecular identifiers (UMIs) assigned to 3245 (human) and 2606 (mouse) genes. To further determine the effect of cell segmentation, we calculated the cumulative proportions of UMIs and area coverage within circular regions that expand progressively at a fixed step size, centered at each segmented cell's centroid. These proportions were calculated relative to the total UMIs detected and the surface area of the S1 chip (see materials and methods). We found that 67.98% of transcripts from all segmented cells were confined to just 5.62% of the total chip area (fig. S1A), suggesting that Stereo-cell enables accurate RNA intensity-based cell segmentation.

To evaluate the lateral RNA diffusion (18), we analyzed data from 10,000 human-mouse mixed cells to compare the radii of estimated actual cells, as determined by DAPI signal distributions, with segmented cells based on captured RNA signal distributions (fig. S1B). The distribution density results revealed that the radius length is moderately greater in segmented cells (peak value, ~15 μm) compared with that of estimated actual cells (peak value, ~10 μm), suggesting a certain degree of lateral RNA diffusion (fig. S1B). To further assess the impact of lateral RNA diffusion on intracellular molecular resolution, we calculated the intercellular center-to-center distance between each cell and its nearest cell (peak value, ~45 μm) (fig. S1C) as well as the proportion of RNA from other species present in each cell (peak value, ~0.027) (fig. S1C), suggesting that lateral RNA diffusion has minimal impact on intracellular RNA quantification under optimized cell distribution density conditions.

To mitigate doublets—where multiple cells are inadvertently profiled as one—Stereo-cell uses imaging-based cell identification. Cells were stained with a variety of dyes, such as DAPI, trypan blue, and 1,1'-Diocetadecyl-3,3,3',3'-tetramethylindocarbocyanine perchlorate (DiI) for imaging (Fig. 1, D and E, and fig. S1D), and regions containing multiple nuclei were excluded (Fig. 1, F and G, and fig. S1, E and F). This reduced the doublet rate from 4.38 to 1.23% for a 5000-cell input (fig. S1G), including the homotypic doublets that could not be identified by scDblFinder (fig. S1H) (21), while yielding 9567 UMIs assigned to 2960 genes (fig. S1I). To assess technical reproducibility, we profiled two additional 5000-cell replicates on the S1 chip (fig. S2, A and B), revealing consistency across various metrics, including doublet rates post-image-based filtering (fig. S1G and fig. S2, C and D), number of UMIs and genes captured (fig. S1I and fig. S2, C and D), and overall correlation (fig. S2E). Experiments using 200- and 2000-cell inputs showed high uniformity in terms of doublet rates following image-based filtering and cell recovery capabilities (fig. S3, A and B, and table S1). However, the 200-cell condition yielded lower median UMIs and gene counts per cell (figs. S1I and S3C), likely owing to reduced library complexity and elevated background, as suggested by down-sampling analysis (fig. S3D).

Stereo-cell enables accurate and high-throughput single-cell transcriptome profiling across a wide range of input conditions, with integrated imaging facilitating robust cell segmentation and doublet exclusion.

Comprehensive profiling of human peripheral blood mononuclear cells

To evaluate Stereo-cell on a heterogeneous primary cell population, we profiled around 10,000 human peripheral blood mononuclear cells (PBMCs) on the S1 chip. DAPI staining was performed to ascertain the position of each attached PBMC (fig. S4A) and to exclude the cell

doublets that contained multiple nuclei within a continuous UMI area (fig. S4B), resulting in 7581 single cells after filtering.

For benchmarking, we integrated Stereo-cell PBMC data with publicly available 10x Chromium PBMC datasets of a similar cell number (see materials and methods). Dimensionality reduction and uniform manifold approximation and projection (UMAP) was performed on mixed cells, revealing that the distribution and gene expression profiles of cells in each annotated cluster were largely in agreement between Stereo-cell and 10x Chromium platforms (Fig. 2, A and B, fig. S4, C and D, and table S2). We further confirmed the high consistency across platforms and reproducibility of technical replicates of Stereo-cell data (Fig. 2C). Of note, when we calculated UMI number per cell, gene number per cell and the proportion of mitochondrial reads using full or down-sampled data that were normalized to 20,000 reads per cell, we observed higher numbers of detected transcripts and genes, yet lower nonspecific capture of mitochondrial reads from data by Stereo-cell compared with that by the 10x Chromium platform, suggesting superior capture efficiency and specificity of Stereo-cell (Fig. 2D and fig. S4E). Down-sampling analysis across multiple depths showed that the relative ranking of gene detection sensitivity between platforms remained stable (fig. S4F).

We also compared key performance parameters across major scRNA-seq platforms from previous studies, including Smart-seq2 (22), CEL-Seq2 (22, 23), Seq-Well (24), BD Rhapsody (25), Microwell-seq (26), 10x Chromium (25), Drop-seq (10, 22), inDrops (11, 22), DNBeLab C4 (27), sciFi-RNA-seq (9), and PIP-seq (13). In addition to showing overall performance comparable to that of mainstream technologies, Stereo-cell offers advantages in technical compatibility with diverse cell types and imaging integration (table S3).

To evaluate clustering performance, we integrated Stereo-cell-generated PBMC data with publicly available datasets from multiple platforms (22, 27) for unsupervised clustering analysis (fig. S5A). Overall, Stereo-cell successfully recovered the abundant cell types in PBMCs. Notably, upon meticulous comparative analysis of captured cell type proportions, our findings revealed that the proportion of monocytes detected by Stereo-cell was 16.79% (Fig. 2E), close to that determined by flow cytometric analysis in PBMC samples (Fig. 2F, 17.1%), yet much lower compared with the 28.31% identified in public datasets from the 10x Chromium platform (Fig. 2E). Similarly, the percentage of other principal cell types within PBMCs, including T cells, B cells, and CD16⁺ monocytes or natural killer (NK) cells captured by Stereo-cell, showed a closer resemblance to those obtained by flow cytometry (Fig. 2, E and F, and fig. S5, B and C).

These results suggest that Stereo-cell achieves accurate, high-resolution profiling of diverse immune populations and captures cell type proportions more representative of ground-truth measurements such as flow cytometry.

Ultrahigh-throughput cell profiling: Identification of rare cell populations

A substantial increase of cell throughput in a single experiment would not only save time and reduce labor costs but also minimize batch effects, thereby alleviating the challenges associated with data integration (9, 28). The flexible manufacture of Stereo-seq chips at different sizes from 0.5 $\times$ 0.5 to 13 $\times$ 13 cm hold the potential to further enhance the cell throughput of Stereo-cell. To demonstrate this potential, we performed Stereo-cell profiling using PBMCs at cell quantities of 7000, 250,000 and 900,000 on the 0.5 $\times$ 0.5– (S0.5), 3 $\times$ 3– (S3) and 6 $\times$ 6–cm (S6) chips, leading to the successful capture number of 5027, 134,902 and 445,467, respectively (Fig. 2, G to I). Of note, the higher cell throughput achieved from the S3 and S6 chips compared with that from the S0.5 and S1 chips allowed for a more detailed classification of cell populations (Fig. 2, A and G to I) while maintaining comparable transcript detection efficiency at the same sequencing depth (fig. S6, A and B). Cell type classification at higher resolution enables

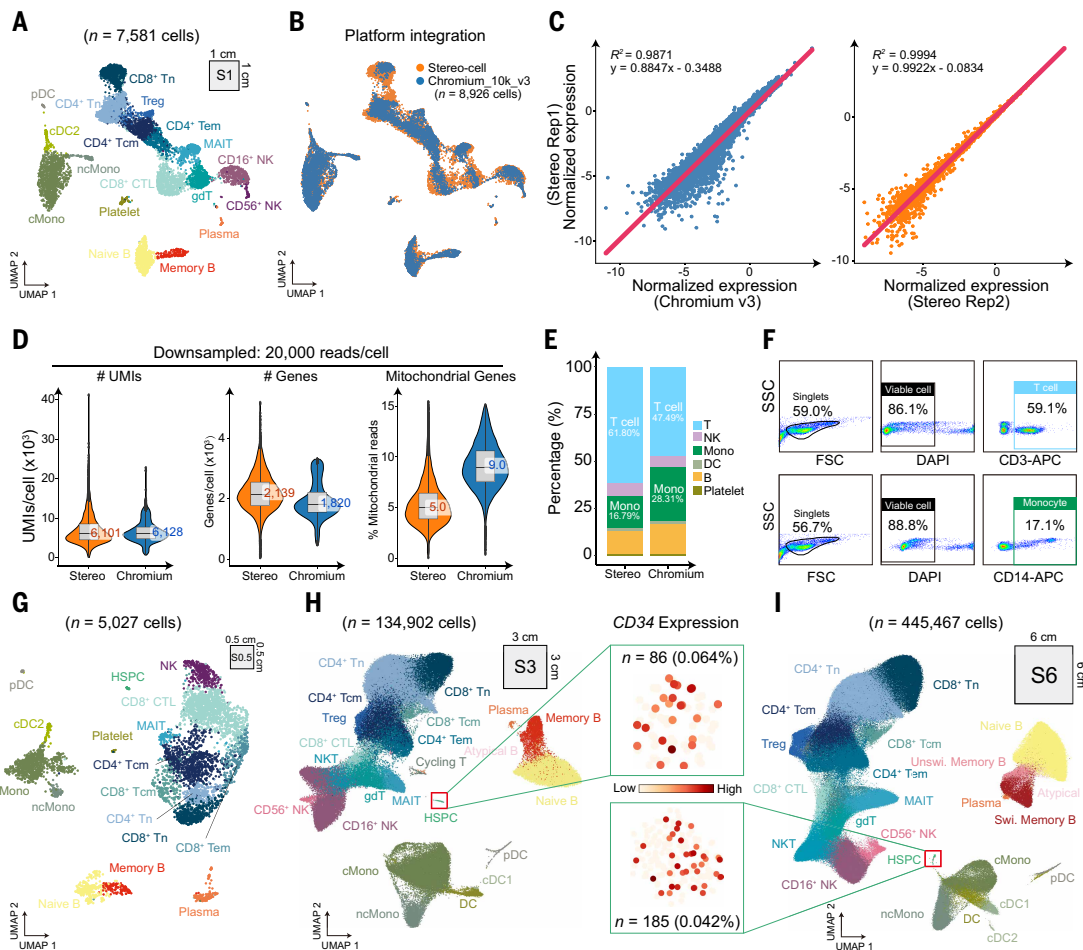


Fig. 2. Stereo-cell enables reliable, accurate, and flexible transcriptional profiling at the single-cell level. (A) UMAP visualization of 7581 human PBMCs applied onto the Stereo-cell S1 chip, with cells colored by cell types. (B) UMAP visualization of 7581 Stereo-cell PBMCs integrated with 8926 10x Chromium PBMCs, with cells colored by platforms. (C) Correlations in average normalized gene expression between platforms (left) and technical replicates of Stereo-cell (right). R^2 , coefficient of determination. (D) Violin plots showing the number of genes, UMIs captured, and the proportion of mitochondrial genes between Stereo-cell and 10x Chromium using down-sampled data normalized to an equivalent number of reads (20,000 reads per cell). In the box plots, the center lines represent the medians, and the upper hinge and lower hinge represent 25% and 75% quantiles, respectively. (E) Proportional representation of principal cell types within PBMCs in Stereo-cell and 10x Chromium. (F) Flow cytometric analysis of scatter profile showing the cell proportion of CD14⁺ monocytes (top) and CD3⁺ T cells (bottom). SSC, side scatter; FSC, forward scatter. (G) UMAP visualization of 5027 human PBMCs applied onto the Stereo-cell S0.5 chip, with cells colored by cell types. (H) (Left) UMAP visualization of 134,902 human PBMCs applied onto the Stereo-cell S3 chip, with cells colored by cell types. (Right) Expression of marker genes of HSPCs (CD34) overlaid on magnified UMAP clusters of human PBMCs applied onto Stereo-cell S3 (top) and S6 (bottom) chips, derived from the region squared in (H) and (I), respectively. (I) UMAP visualization of 445,467 human PBMCs applied onto the Stereo-cell S6 chip, with cells colored by cell types.

Stereo-cell to detect rare yet important cell types, such as human hematopoietic stem and progenitor cells (HSPCs, ~0.05%), discernible on the S3 and S6 chips (Fig. 2, H and I). These results demonstrate the capability of Stereo-cell to handle ultrahigh throughput and to characterize rare and elusive cell populations, achieving a reasonable identification of cell clustering.

Harmonizing multiplex immunofluorescence within the Stereo-cell framework

Single-cell transcriptome data alone may be insufficient to fully resolve cellular phenotypes and functional states (2). Stereo-cell's planar chip format enables integration with imaging-based assays, including multiplex immunofluorescence (mIF), allowing concurrent measurement of transcriptomes and protein markers at single-cell resolution.

To test this integration, we developed a workflow combining Stereo-cell with mIF staining (see materials and methods). Following cell fixation, we utilized DAPI and wheat germ agglutinin (WGA) conjugate

to stain the cell nucleus and cell membrane, respectively. For cell type-specific proteins, we used antibodies labeled with distinct fluorophores (anti-CD3-FITC and anti-CD45RA-TRIFITC) to specifically target and visualize the CD3 and CD45RA proteins within PBMCs (fig. S6C, top). Standard Stereo-cell procedure was then followed to generate transcriptomic data of stained cells (mIF replicates 1 and 2). The average numbers of detected transcripts per cell for mIF and Stereo-cell samples were comparable to those for standard Stereo-cell samples at the same sequencing depth (fig. S6, A and B), and highly consistent cell-type classification was observed between the two methods (fig. S6D), suggesting that mIF introduces little bias of transcript capture into Stereo-cell.

Protein signals were spatially concordant with transcriptome-based annotations (fig. S6, C to E). CD3 protein was localized within transcriptomically defined T cells (fig. S6C, left, and fig. S6E, left) (29), whereas CD45RA protein was expressed in B cells, NK cells, and naïve T cells (fig. S6C, middle, and fig. S6E, middle) (30). In addition, we

noted intensified WGA signals in monocytes (fig. S6C, right, and fig. S6E, right), which may be attributable to their extensive plasma membrane surface area labeled by WGA, in line with previous flow cytometry findings (37).

Altogether, these results indicate that Stereo-cell contains expandable capacity for achieving fluorescent signal acquisition of cell staining and accurate cell characterization.

Simultaneous transcriptome and epitope profiling with Stereo-cell-CITE

To expand the multiomics capability of Stereo-cell, we adopted the cellular indexing of transcriptomes and epitopes by sequencing (CITE-seq) strategy and developed Stereo-cell-CITE to simultaneously capture transcripts and oligonucleotide-barcode antibodies specifically targeting cell surface proteins (Fig. 3A, left) (see materials and methods) (32). PBMCs were incubated with a panel of 154 well-characterized antihuman immune cell antibodies and 9 isotype controls, then profiled at a density of 10,000 cells per S1 chip under unstimulated control or costimulation with Phorbol 12-myristate 13-acetate (PMA) and ionomycin (Fig. 3A, right) (see materials and methods). Following cell permeabilization, cellular mRNAs and antibody-derived tags (ADTs) underwent simultaneous annealing through their 3' polyadenylate [poly(A)] tails to DNB probes containing oligo-dT, with subsequent independent complementary DNA (cDNA) and ADT library construction and sequencing (see materials and methods). While maintaining robust transcript capture efficiency (fig. S7, A and B), the resultant RNA signal distributions demonstrated notable colocalization with protein (ADT) signals within individual cells (Fig. 3B), enabling simultaneous transcriptomic and proteomic single-cell analysis. To validate the consistency of marker protein and transcript identification for cellular classification, we analyzed the expression distribution of protein signals in the integrated cell atlas of all samples derived from gene expression (Fig. 3, C and D, and table S4). Expression of surface proteins—such as CD13, CD14, and CD88 (myeloid cell clusters) (33–35); CD4, CD8, and KLRG1 (T or NK cells) (36–38); and CD19 and IgD (B cells) (39, 40)—matched corresponding transcriptomic clusters (Fig. 3, C and D).

Integrating single-cell multiomics data from Stereo-cell-CITE provided a finer resolution of cellular heterogeneity, enabling more nuanced classification of cell types and functional states. For example, a T cell subcluster characterized by elevated surface expression of CD112, an adhesion molecule involved in immune cell interaction (41), was distinguished only when protein signals were included (Fig. 3, E and F, left, and fig. S7C). Similarly, elevated CD103 expression identified a subset of tissue-resident immune cells (42, 43), which are not readily resolved from transcript data alone (Fig. 3, E and F, right, and fig. S7D).

PMA and ionomycin costimulation elicit a multifaceted cascade of cellular responses in PBMCs, encompassing cytokine production, cellular activation, and proliferative dynamics (44–47). Our results indicate that the stimulation led to a decrease in the proportion of monocytes and dendritic cells, accompanied by a concomitant increase in activated immune cell clusters within the PBMC population (Fig. 3G and fig. S7E). This cellular population reduction potentially stems from myeloid cell differentiation induced by up-regulated cytokines (e.g., *IL-2* and *INFγ*) and chemokines (e.g., *CXCL2*, *CXCL3*, and *CXCL8*) under stimulatory conditions (fig. S7, F and G) (48, 49). Notably, CD62L and CD16 protein expression decreased across diverse immune cell types after stimulation (Fig. 3H), consistent with prior reports of their down-regulation during activation (50, 51).

To elucidate molecular interactions, we computed Pearson's correlation coefficient between mRNA and ADT expressions, constructing control- and stimulus-specific protein-RNA coexpression networks (Fig. 3I and fig. S7H). Within the stimulus-specific network, the gene *KLRD1* and its encoded protein CD94 emerged as central nodes, exhibiting extensive interconnections with critical immune regulatory

molecules, including CD56, CD244, CD28, and CD27 (Fig. 3I), implicating their involvement in NK cell regulatory pathways (52).

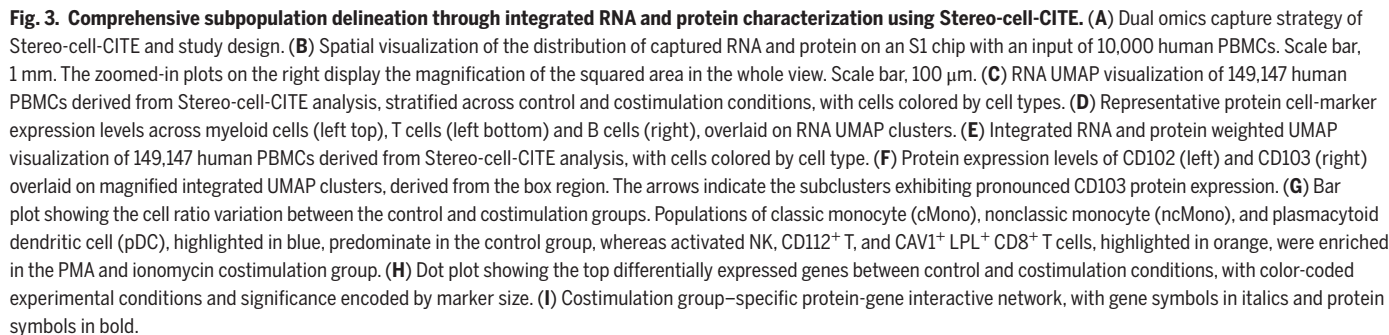
These findings highlight the added value provided by multiomics analysis of Stereo-cell-CITE in exploring subtle differences in cellular characteristics under varying conditions that remain imperceptible through transcriptomic analysis alone.

In situ capture of transcriptome from cultured cells utilizing Stereo-cell

Another major advantage of the silicon chip with flat PLL-coating surface is that it supports cell culture, thus allowing for the direct recording of the dynamic cell states at the transcriptome level by Stereo-cell. To validate the stability and integrity of the DNBs on the chip under cell culture conditions, we applied 18 types of oligonucleotides with UMI barcodes and poly(A) tails across a concentration gradient to chips that had undergone 64 hours of cell culture followed by thorough cell digestion and removal (fig. S8A) (see materials and methods). Both cell-covered regions and four progressively distant uncovered regions, estimated by the distribution of nuclear imaging signals from DAPI staining (fig. S8A, right) (see materials and methods), demonstrated consistency in the capture efficiency across all oligonucleotide concentrations (fig. S8, B and C). To further assess potential influence of the culture environment and cell dissociation on cells, we compared transcriptomes of NIH-3T3 fibroblasts collected under three conditions: (i) profiling for single-cell suspensions obtained by dissociation after 64 hours of culture in a dish, (ii) profiling for single-cell suspensions obtained by dissociation after 64 hours of culture on the DNB chip, and (iii) in situ profiling for cells cultured on the chip for 64 hours (see materials and methods). The high consistency and correlation between the two cell dissociation groups indicated that the substrate DNA on the chip surface did not alter cellular gene expression (fig. S8, D and E, and table S5). Meanwhile, the in situ capture group exhibited high expression of genes related to extracellular matrix organization, cell migration, and cell-substrate adhesion (fig. S8F) and was distinguished from the cell dissociation groups, which showed high expression of genes implicated in apoptotic processes (fig. S8G). This suggests that in situ capture and sequencing of cultured cells by Stereo-cell facilitate the investigation of changes in cellular gene expression due to cell-to-cell contact and cell migration while excluding the effects of artificial gene alteration caused by cell dissociation.

As a proof of concept, we conducted further analysis on the group of cultured 1000 NIH-3T3 cells on the chip for in situ transcript capture (Fig. 4A). Clustering revealed 11 cell states, including distinct cell cycle phases and transcriptional profiles (Fig. 4, B to D, and fig. S9A). Notably, one distinct population, including clusters 0, 6, and 9, exhibited significantly lower numbers of genes (average = 272, $P < 0.01$) and DNB spots (average = 1724, $P < 0.01$) per designated area compared with those of other cell clusters (fig. S9B). These clusters corresponded to small structures localized near fibroblast dendrites, visualized by DiI staining (Fig. 4, E and F). Notably, these clusters expressed extracellular vesicle (EV)-associated genes, such as *Gphn*, *Cops2*, and *Cops5* (Fig. 4, C and D) (53–55), suggesting that they represent EVs. Using SEVtrac, an EV identification and analysis algorithm based on gene set enrichment and expectation-maximization modeling (see materials and methods) (56), we confirmed that these clusters displayed transcriptomic signatures consistent with EVs (Fig. 4G). In addition, the mIF assay results using antibodies for EV-specific positive (SYN) (57) and negative (Calnexin) (58) markers confirmed the presence of EVs in the adjacent regions of the NIH-3T3 cells cultured on the chip (Fig. 4H) (see materials and methods). Thus, these results suggest that Stereo-cell is capable of capturing transcripts contained in cell-derived membranous structures.

To examine responses to stimulation, we treated NIH-3T3 cells with lipopolysaccharide (LPS) for 4 hours before in situ profiling (Fig. 4A).



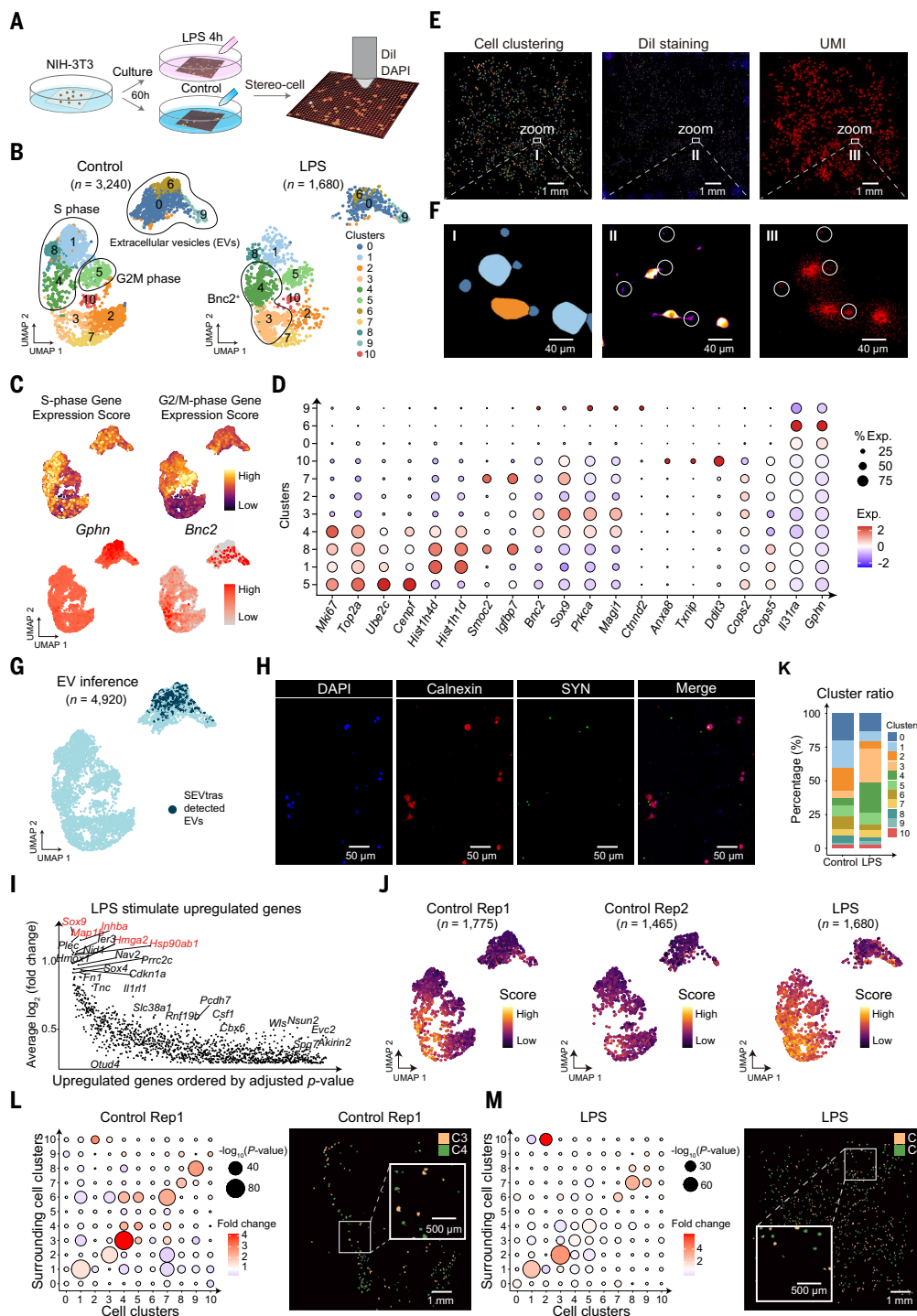


Fig. 4. Stereo-cell enables in situ sequencing for cultured cells. (A) Experimental outline of in situ sequencing for cultured NIH-3T3 cells. (B) UMAP visualization of NIH-3T3 cells directly cultured on the S1 chips under control cultured condition (left) and LPS stimulation (right). Cells are colored by cell clusters. (C) (Top) Gene expression score of S and G2/M cell cycle phases overlaid on UMAP clusters. (Bottom) Expression of representative marker genes *Gphn* and *Bnc2* overlaid on UMAP clusters. (D) A dotted heatmap showing the scaled expression of representative marker genes for each cell cluster. (E) Spatial visualization of the distribution of cell clusters (left), Dil staining signals (middle), and captured transcripts (right) on an S1 chip with cultured cells. The cell clusters are color coded as they are in (B). Scale bar, 1 mm. (F) Magnification showing the spatial distribution of segmented cells colored by clusters (left), Dil staining signals (middle), and captured transcripts signals (right) from the region squared in (E). The white circular frame indicates EV cluster 0. Scale bar, 40 μ m. (G) SEVTras-inferred extracellular vesicles, colored in dark blue, overlaid on UMAP clusters. (H) mIF assay of on-chip cultured NIH-3T3 cells, revealing signal distribution of Calnexin (red), SYN (green), and nuclear counterstaining with DAPI (blue). The composite merged image is shown to the right. Scale bar, 50 μ m. (I) Scatter plots of the expression fold changes of up-regulated genes following LPS stimulation. (J) Activity of the TGF- β signaling pathway overlaid on UMAP clusters of replicates 1 (left) and 2 (middle) under control cultured condition as well as on the LPS stimulation group (right). (K) Proportional representation of cell cluster ratio between the control cultured group and the LPS stimulation group. (L and M) (Left) A dotted heatmap showing the frequency of surrounding cell clusters for each cell cluster within the control replicate 1 group (L) and the LPS stimulation group (M). (Right) Spatial visualization of the distribution of cells belonging to clusters 3 and 4. Magnification showing details from the region squared on the chips. Scale bars, 1 mm and 500 μ m.

Genes functionally associated with the transforming growth factor β (TGF- β) signaling pathway, such as *Sox9*, *Map1b*, *Inhba*, *Hsp90ab1*, and *Hmga2*, were among the top up-regulated ones following LPS treatment (Fig. 4I and table S6). We also noted that, compared with that of the control group, the overall intensity of the TGF- β signaling pathway in the LPS-stimulated group was enhanced (Fig. 4J). A previous study has indicated that a key function of the TGF- β pathway is to regulate the migration of NIH-3T3 cells (59), suggesting that fibroblasts collectively exhibited migration characteristics following LPS introduction.

The integrated clustering results revealed enrichment of cell populations in clusters 3 and 4 (Fig. 4, B and K, and fig. S9C) characterized by *Bnc2*, *Sox9*, and *Prkca* (Fig. 4, C and D), markers that were indicative of fibrotic activation induced by LPS in fibroblast cells (60–62). These cells tended to aggregate (Fig. 4, L and M, and fig. S9D), exhibiting stronger intercellular interactions compared with other cell clusters (fig. S9E, left and middle). The interactions between most cell clusters and fibrosis-associated clusters (clusters 3 and 4) also intensified following LPS stimulation (fig. S9E, right), implying a link between the fibrotic process and increased cellular interactions among fibroblasts. Furthermore, we observed that cluster 3 and 4 cells engaged in communication through potential signaling pathways, such as cadherin and collagen pathways (fig. S9F). Conversely, a reduction in EV cluster 6 (marked by *Il31ra*) was observed after LPS stimulation (Fig. 4, B, D, and K, and fig. S9C), suggesting a potential role for EV down-regulation in modulating fibrosis, as *Il31ra* deficiency has been associated with reduced fibrotic response (63).

Together, these findings underscore the capability of Stereo-cell to detect in situ cellular interactions and microenvironmental changes.

Capturing and sequencing large-sized microstructures using Stereo-cell

Multinucleated skeletal myofibers present a specific challenge for single-cell transcriptomic analysis because of their size and structural complexity. Microfluidic-based methods are incompatible with such large structures, and although single-nucleus RNA-seq can capture nuclear diversity, it loses spatial context (64, 65). Stereo-cell, with no inherent size limitations, enables transcriptome capture while preserving the spatial position of individual nuclei, providing a more informative approach for analyzing myofibers.

To demonstrate this, we extracted skeletal myofibers from the soleus and extensor digitorum longus (EDL) muscles of *Mus musculus* for Stereo-cell analysis (Fig. 5A) (see materials and methods). A total of 17 myofibers were captured on two S1 chips, followed by library construction and sequencing. The incorporation of nuclear staining and imaging into the workflow facilitated spatial mapping of nuclear distributions within individual myofibers (fig. S10A). By using a *K* means-based region of interest (ROI) parcellation methodology, we defined analysis bins with an approximately 39- μ m diameter (bin 80, 80 $\times$ 80 DNB) (fig. S10B) (see materials and methods) and performed unsupervised clustering of ROIs across all fibers (Fig. 5B; fig. S10, C and D; and table S7) (see materials and methods). Consistent with previous report, the slow-twitch soleus muscle comprised type I (slow) myofibers predominantly expressing *Myh2* and *Myh7* (Fig. 5, B, left, and C, top) (66) but also included types with other expression characteristics, such as type II (fast), including clusters 0, 1, and 2, marked by *Myh1* expression, and a type potentially associated with muscle repair and regeneration, comprising cluster 5 with enriched mRNA expression of *Sfrp5*, an endogenous inhibitor of Wnt signaling crucial for maintaining muscle homeostasis (Fig. 5, B, left, and C, top, and fig. S10C) (67–69). By contrast, EDL myofibers predominantly expressed type IIb (fast) marker *Myh4*, revealing functionally heterogeneous clusters with distinct molecular signatures (Fig. 5, B, right, and C, bottom). For example, the elevated expression of *Eef1a1* in cluster 6 and *Dmd* in cluster 0 suggests that these two distinct groups of ROIs

may facilitate muscle repair and protect muscle fibers from damage through supporting protein synthesis and providing structural stability to muscle cell membranes by anchoring the cytoskeleton, respectively (Fig. 5, B, right, and C, bottom, and fig. S10D) (70, 71). Marker genes *Col22a1* of cluster 4 and *Chrna1* of cluster 5 are particularly expressed at the myotendinous junction (MTJ) and neuromuscular junction (NMJ), respectively (Fig. 5, B, right, and C, bottom, and fig. S10D) (72, 73), suggesting that these two types of ROIs play a role in the connection between myofibers and other different tissues.

More notably, Stereo-cell data enabled us to pinpoint the exact location of each ROI within the myofibers (Fig. 5, D and E). In the soleus myofibers, each muscle fiber was predominantly composed of one class of myofiber ROI, which could therefore be classified into distinct types accordingly, such as type II (fast) with *Myh1* expression, type I (slow) with *Myh2* or *Myh7* expression, and a homeostasis-like type with *Sfrp5* expression (Fig. 5D). In EDL myofibers, these ROIs with different functions exhibited distinct spatial distribution characteristics (Fig. 5E). Notably, the ROIs with high expression of *Col22a1* were distributed at both ends of the fiber, whereas those with high expression of *Chrna1* were located in the middle of the fiber (Fig. 5E), consistent with the spatial distribution characteristics of MTJ and NMJ. We found that, in type II (fast) myofibers, *Myh1* preferred a distribution pattern at both ends of the fiber, similar to in MTJ, compared with the widespread distribution of *Myh4* (Fig. 5E). Moreover, RNAscope results showed a strong concordance between the RNA staining patterns and the gene expression profiles obtained from Stereo-cell of *Myh1* and *Myh4* (Fig. 5F) (see materials and methods), indicating the accuracy of Stereo-cell in uncovering spatial heterogeneity in muscle fibers.

These findings of spatial heterogeneity in EDL muscle fibers prompted us to further search for spatially covarying gene modules with our Stereo-cell dataset by the Hotspot algorithm (Fig. 5G, fig. S10E, and table S8) (see materials and methods) (74). For example, among nine modules identified within the representative EDL myofibers, module 2 (M2) exhibited the MTJ pattern (Fig. 5G). Aside from previously identified characteristic genes, such as *Myh1* and *Col22a1*, we detected genes such as *Lama2* and *Fgf6* in M2 (Fig. 5G), the expression patterns of which, in myofibers, were in line with those of previous studies (75–77). Moreover, we succeeded in identifying module 1 (M1), which was sparsely distributed in the middle of the myofibers (Fig. 5G). M1 can be characterized as a spatial module associated with the NMJ, as it comprises genes indicative of the neuromuscular junction, including *Chrna1*, *Phldb2*, and *Ncam1* (Fig. 5G) (78, 79).

Our data demonstrate that Stereo-cell can capture and sequence large-size microstructures and robustly distinguish spatial segments in microstructures.

High-throughput characterization of a large-sized cell: the oocyte

We next examined whether Stereo-cell is suitable for analyzing cell samples of large sizes. As the largest cell type in mammals, the oocyte has been studied at low throughput by tube-based single-cell sequencing approaches following manual picking or fluorescence-activated cell sorting (FACS) (80, 81). In this study, we harvested 719 oocytes from the ovaries of 3- and 8-week-old C57BL/6J mice treated with pregnant mare serum gonadotropin (PMSG) and placed them onto five S1 chips (three chips for 3-week-old mice and two chips for 8-week-old mice) (Fig. 6A and fig. S11A). On average, Stereo-cell captured 8972 genes per oocyte (fig. S11B). Unsupervised clustering identified seven subtypes corresponding to four key maturation stages (Fig. 6B, left and right top, and table S9). Clusters 1, 2, and 6 expressed canonical growing-stage markers *Bmp15* and *Gdf9* (82, 83). Cluster 0 was associated with germinal vesicle breakdown (GVBD) stages, and clusters 3 and 4 were linked to germinal vesicle (GV), as indicated by the expression of GV-specific markers, *H1foo* (84). Cells in cluster 5 appeared to be at meiosis (M) stage and expressed *Dcp2* (Fig. 6C) (85). The identified GV stage aligned with prior classifications (Fig. 6B, right

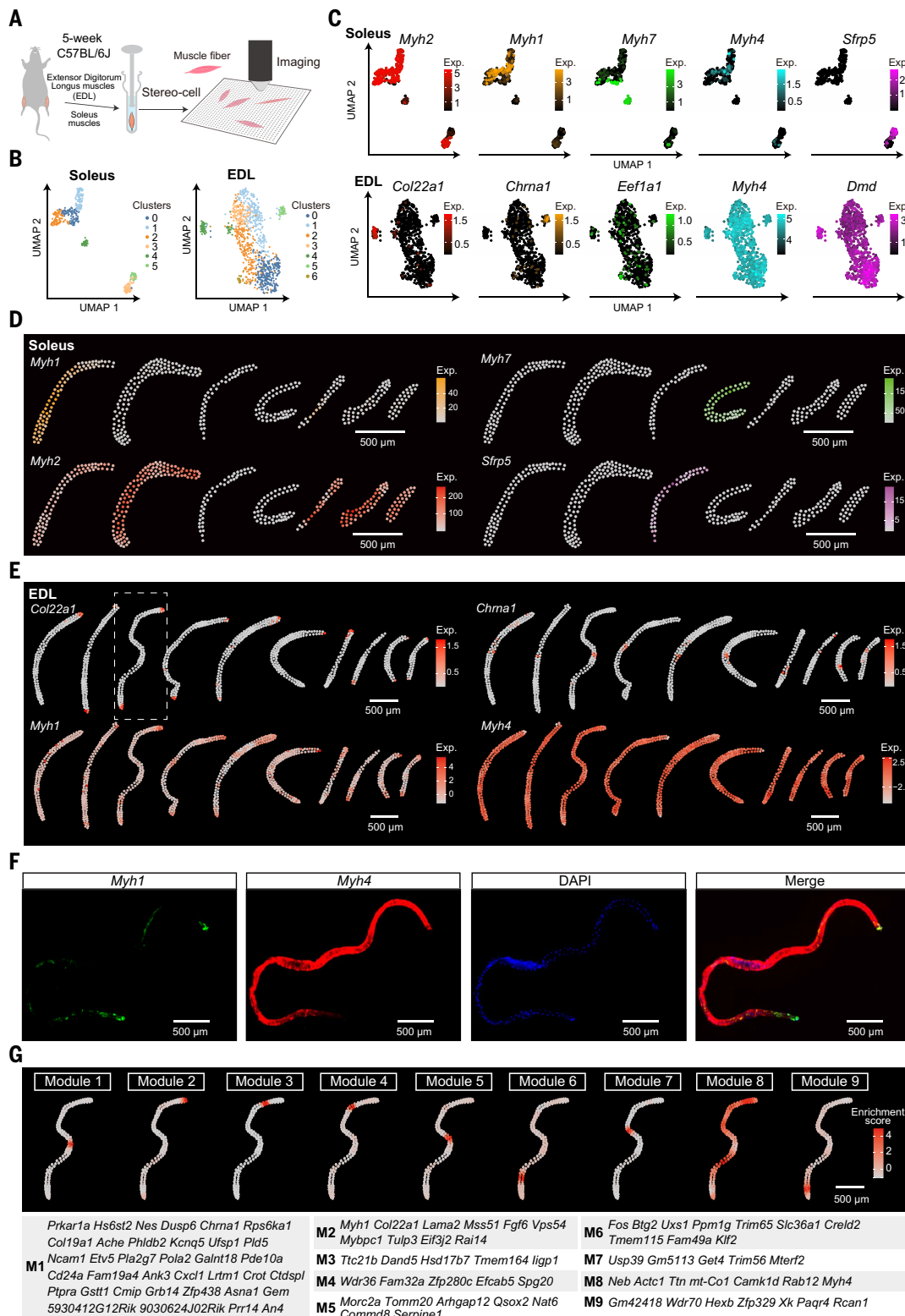


Fig. 5. Capturing and sequencing of the large-sized microstructure, multinucleated skeletal myofiber, using Stereo-cell. (A) Experimental outline of Stereo-cell sequencing for isolated skeletal soleus and EDL myofibers. (B) UMAP visualization of ROIs within all captured soleus (left) and EDL myofibers (right), with ROIs colored by cluster. (C) Expression of representative cluster differentially expressed genes overlaid on the UMAP clusters of soleus (top) and EDL (bottom). (D) Spatial expression patterns of *Myh1* (orange), *Myh2* (red), *Myh7* (green), and *Sfrp5* (purple) within representative soleus myofibers. Scale bar, 500 μ m. (E) Spatial expression patterns of *Col22a1* (top left), *Chrna1* (top right), *Myh1* (bottom left), and *Myh4* (bottom right) within representative EDL myofibers. Scale bar, 500 μ m. (F) RNAscope assay of *Myh1* (green), *Myh4* (red), and DAPI (blue) staining within a representative EDL myofiber. The composite merged image is shown to the right. Scale bar, 500 μ m. (G) (Top) Spatial expression patterns of all gene modules detected in the representative EDL myofiber indicated by the dashed box in (E). (Bottom) Gene members of all gene modules. Scale bar, 500 μ m.

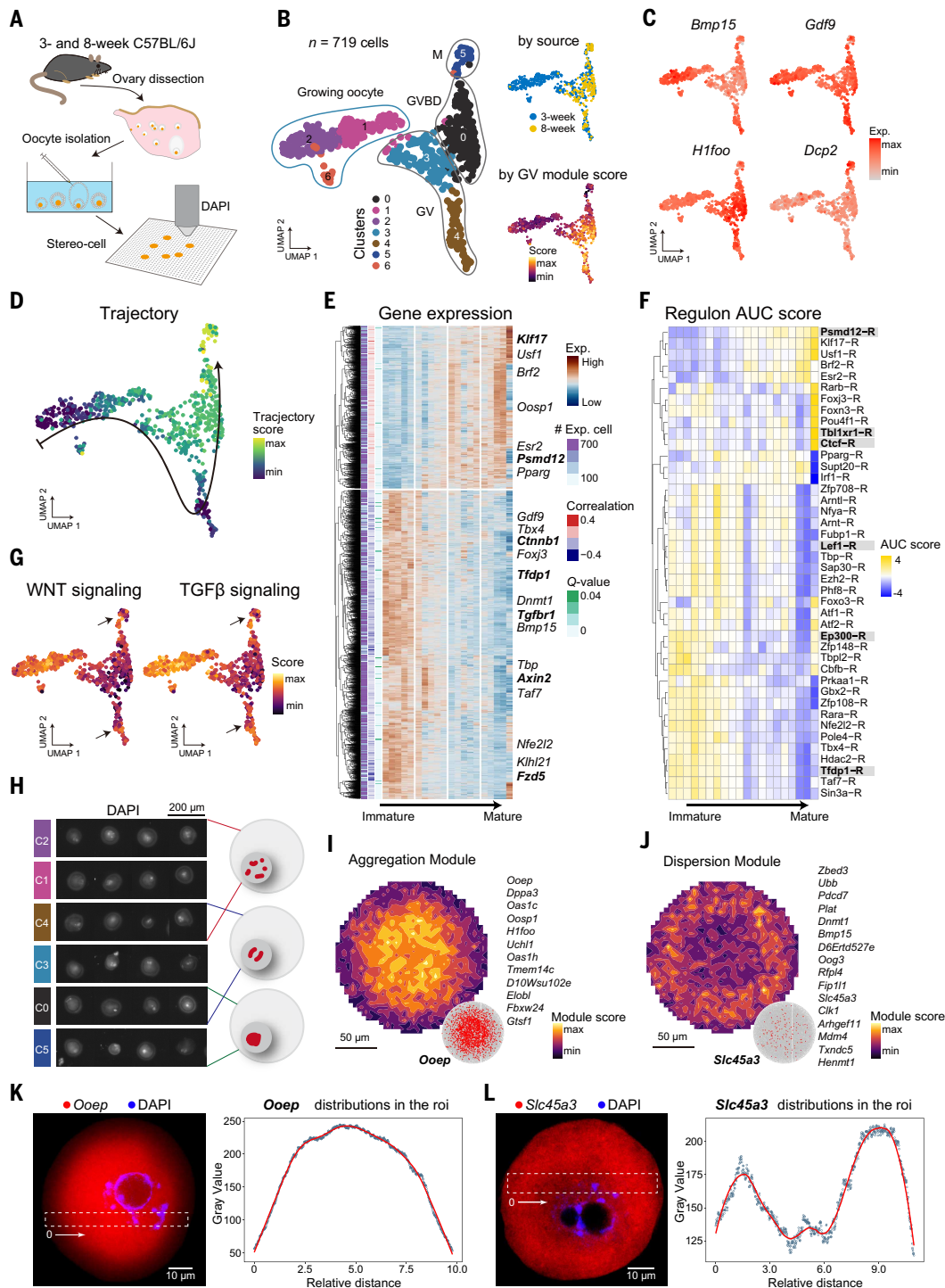


Fig. 6. Capturing and sequencing of the oocyte, a large-sized cell, using Stereo-cell. (A) Experimental outline of isolation and Stereo-cell sequencing for oocyte from C57BL/6J mice at 3 and 8 weeks of age. (B) UMAP visualization of all 719 oocytes, with cells colored by cluster (left), oocyte source (right top), and GV module score (right bottom). GV module scores were calculated based on the method described by Hu *et al.* (86). (C) Expression of marker genes of the growing (*Bmp15*, *Gdf9*), GV (*H1foo*), and M (*Dcp2*) stages, overlaid on UMAP clusters of oocytes. (D) The cell trajectories are visualized with UMAP [as in (B)], with cells colored by pseudotime. The arrow corresponds to pseudotime directions learned by Monocle 3. (E) Heatmap illustrating the scaled gene expression along pseudotime. (F) Heatmap showing the normalized activity score (AUC) of the indicated regulons along pseudotime. (G) Activity of the gene module per the Wnt (left) and TGF- β (left) signaling pathways overlaid on UMAP clusters of oocytes. The arrows indicate high-activity areas. (H) Representative DAPI nuclei staining images showing the chromatin morphology of each cluster, with accompanying diagrams on the right summarizing the chromatin morphology characteristic across different clusters. Scale bar, 200 μ m. (I and J) (Left) Spatial visualization of module scores for aggregation gene module (I) and dispersion gene module (J) as well as spatial expression patterns of representative genes *Ooep* (I) and *Slc45a3* (J). (Right) Representative genes in aggregation gene module (I) and dispersion gene module (J). Scale bar, 50 μ m. (K and L) (Left) smRNA-FISH results of a representative oocyte for *Ooep* (K) and *Slc45a3* (L), with nuclear counterstaining using DAPI. (Right) Quantitative analysis of the *Ooep* (K) and *Slc45a3* (L) smRNA-FISH signal intensities within the dashed squared ROIs from the left panels. Scale bar, 10 μ m.

bottom) (86), further supporting the accurate capture of transcripts in oocytes by high-throughput Stereo-cell.

To assess gene expression dynamics in developing oocyte, we applied Monocle 3 (28) and diffusion maps (87) to our data, both showing that the ordering of oocytes along the pseudotime maturation path well recapitulated the actual oocyte maturation process (Fig. 6, B and D, and fig. S11, C to E). To further probe the regulatory mechanisms underlying oocyte maturation, we utilized Monocle 3 (28) and SCENIC (88) to respectively dissect differentially expressed genes (Fig. 6E) and predict key transcription factors (TFs) (Fig. 6F) with high regulatory activity along the pseudotime transition, leading to the identification of a series of potential regulatory factors in the canonical Wnt signaling pathway, including the TF *Left1* (fig. S11F) (89), key downstream effector *Ctnnb1* (fig. S11G, left) (90), receptor *Fzd5* (fig. S11G, middle) (91), and key negative regulator *Axin2* (fig. S11G, right) (92). The Wnt signaling pathway is one of the major pathways in maintaining the developmental competence of oocytes (93). Our findings show that the Wnt signaling pathway is highly active during the transition from the growing stage to the GV stage and from the GVBD stage to the M stage (Fig. 6G, left). We also discovered other signaling pathways, TFs, and downstream target genes, such as the TGF- β signaling pathway (Fig. 6G, right) and TFs *Psmid12* and *Tbl1xr1*, with potential target genes that may play roles in stage transition (fig. S11, H and I).

Oocyte maturation is accompanied by chromatin remodeling (94). We observed increased activity of chromatin regulators *Ep300* and *Tfdp1* (fig. S11, J and K, left) (95), whose targets were enriched in pathways related to cell morphogenesis and the Wnt signaling pathway (fig. S11, J and K, right). Nuclear staining revealed a progression from diffuse chromatin in growing oocytes to condensed chromatin at the GVBD and M stages (Fig. 6H), consistent with known nuclear transitions from nonsurrounded nucleolus to surrounded nucleolus and GVBD (94). In addition to the expanded capabilities of cell imaging, the nanometer spatial resolution of DNB chips also endows Stereo-cell with the potential for subcellular RNA localization detection. Using Hotspot (74), we identified spatial gene modules within individual oocyte (see materials and methods). Two notable modules exhibited spatially distinct aggregation and dispersion patterns (Fig. 6, I and J). The representative genes within these two modules, such as *Ooep* and *Slc45a3*, exhibited a strong correlation between their region-specific distributions, as determined by Stereo-cell and validated through single-molecule RNA fluorescence in situ hybridization (smRNA-FISH) (Fig. 6, K and L, and fig. S12, A and B).

Taken together, Stereo-cell is capable of detecting molecules potentially regulating the maturation, chromatin morphological characteristics, and subcellular gene expression localization for large-sized oocytes with high throughput.

Discussion

Innovations based on distinct principles in cell and transcript capture are needed to break the technological constraints of current single-cell sequencing strategies so as to improve the scale of detected cell numbers, prevent capture bias of cell types, and enable profiling of cells with special structures and sizes. The DNB chip was initially designed to profile spatial transcriptomes from tissue sections (18). Owing to its inherent advantages, such as superior spatial resolution, scalable field of view, and encapsulation-free cell and transcript capture capability, the DNB chip can be readily adapted to sequence a population of individual cells and overcome aforementioned long-standing challenges. In this work, we demonstrate that Stereo-cell, based on this chip architecture, enables efficient, scalable, and unbiased transcriptome profiling across diverse cell types and sizes, from small immune cells to large oocytes and multinucleated myofibers.

Stereo-cell can accommodate from low-input (around 200) to high-input cell numbers (around 1 million) while maintaining an unbiased cell type capture. Furthermore, by integrating multiomics information,

Stereo-cell can further explore rare cell types that are challenging to identify using single-omics data alone. These advancements in cell throughput and multiomics integration contribute to the efficiency and scalability of single-cell sequencing studies. More notably, the expanded throughput and a rare cell detection capability offered by Stereo-cell empowers the potential translation of single-cell sequencing into clinical practice. For example, Stereo-cell not only enables sequencing of precious biopsy samples with a small number of cells, but it also may facilitate the capture and identification of rare circulating tumor cells from peripheral blood and ascitic fluid of cancer patients through ultrahigh-throughput cell detection.

By integrating cell staining and imaging with Stereo-cell, it becomes possible to directly visualize the cellular morphology and the spatial distribution of multiple proteins or markers across individual cells while capturing transcripts. When cells were cultured directly on the surface of the chip, the capabilities of Stereo-cell transcend imaging-based detection, enabling the capture and association of cellular morphology, characteristics of cell-cell interactions, and extracellular transcriptomes from microenvironment. This comprehensive integration of cellular and extracellular information with gene expression profiles facilitates a more precise and thorough correlation of cellular phenotypes with various physiological and pathological states.

Our findings demonstrate that Stereo-cell effectively captures and sequences oocytes, highlighting the unbiased cell-size capture capability of Stereo-cell and its great potential for capturing and sequencing other large-sized cells, such as hepatocytes, cardiomyocytes, adipocytes, and even protists, which are typically challenging for conventional high-throughput single-cell platforms. Moreover, with the ability to detect subcellular spatial localization in large-sized cells, future research could investigate their dynamic subcellular expression patterns during physiological and pathological processes.

Despite these strengths, Stereo-cell currently faces several technical challenges. First, chip fabrication and data analysis remain more complex compared with those of mainstream scRNA-seq methodologies. Whereas commercial chips and established automated data analysis platforms (96) allow the experimental focus to shift primarily to the relatively less-time-consuming library construction and sequencing processes, Stereo-cell demands continued advancement in large-scale chip and reagent production alongside algorithmic process optimization. Second, the encapsulation-free design increases susceptibility to lateral RNA diffusion, which, although minimal under optimal cell density, may still contribute to low-level transcript contamination across adjacent cells. Future improvements in cell permeabilization chemistry and capture kinetics could mitigate this effect. Third, transcript detection is reduced under ultralow cell input conditions owing to decreased library complexity. This may be addressed through strategies such as sequencing depth optimization, background signal blocking, multiplexed cell labeling, and the use of spike-in controls. Lastly, there are additional technical limitations, such as its capacity to capture only the transcriptome and selected protein factors, and it is unable to capture other omics components, such as genomic DNA. Furthermore, there is a lack of imaging protocols specifically designed for larger-sized chips, such as S3 and S6 chips. The library preparation process has not yet been fully automated, which could streamline and enhance the efficiency of the workflow. These challenges present opportunities for further development and refinement of the Stereo-cell technique, which holds the potential to render this method more automated, cost effective, and user friendly.

Materials and methods

Cell lines

HEK293T and NIH-3T3 cell lines were obtained from American Type Culture Collection (ATCC) and cultured in the complete growth medium with 90% Dulbecco's Modified Eagle medium (DMEM, GIBCO, #11965092), 10% Hi-Fetal Bovine Serum (Hi-FBS, Biological Industries,

#04-001-IACS), and supplemented with 1x penicillin-streptomycin (GIBCO, #15140122). Cells were grown to a confluence of 50-60%. Prior to use in the experiments, the cell lines were treated with 25% Trypsin-EDTA (Biosharp, #BL512A) for 3 min and then quenched with an equal volume of complete growth medium.

PBMCs

All experiments were approved by the Institutional Review Boards (IRBs) at BGI-Shenzhen (approval number: BGI-IRB24026), and written informed consent was received from all participants prior to the study. Human PBMCs were obtained from four self-identified Asian female donors between 20 and 40 years of age (Shenzhen, China) and resuspended in a cryopreservation solution consisting of 90% Hi-FBS and 10% Dimethyl Sulfoxide (DMSO, Sigma, #D2650). The cells were then cryopreserved in cryotubes (Corning, #430488) then stored at -80°C. The PBMCs were thawed in a 37°C water bath for rapid recovery prior to use in the experiments.

Single-cell suspension preparation

For HEK293T, NIH-3T3, and PBMCs, the single-cell suspensions were prepared as following steps: the suspensions were washed twice with 1x phosphate buffered saline (PBS, Meilunbio, #MA0015) supplemented with 0.04% bovine serum albumin (BSA, Sigma, #A8022-50G) and resuspended, then strained through a 40 µm cell strainer (Falcon, #352340). Cell viability was assessed using trypan blue staining (Sigma, #T8154-20ML), and the single-cell suspension was counted. Subsequently, aliquots were prepared at a final concentration of 500 cells/µL. To evaluate the collision rate, we mixed the HEK293T and NIH-3T3 cells at a 1:1 ratio as input sample.

All animal procedures were performed in compliance with guidelines for the care and use of animals and were approved by the Institutional Animal Care and Use Committee at the Guangzhou Institutes of Biomedicine and Health (approval number: IACUC2023086). For collection of mouse oocytes, female C57BL/6J mice (3- and 8-week-old) were purchased from QIZHEN Laboratory Animal Technology CO., LTD. (Hangzhou, China). Additionally, the 8-week-old C57 mice were administered with 10 IU of PMSG (ProSpec, #HOR-272) for 46 hours. These mice were humanely euthanized, their ovaries were harvested, and oocytes were isolated as previously described (97). Briefly, the dissociation of ovaries was performed using 1x TrypLE Express Enzyme (GIBCO, #12605010) and 2 mg/mL collagenase IV (BBI, #A004186-0001), maintained at 37°C for 20 min. This step was followed by releasing the oocytes to a 37°C pre-heated M2 medium (Sigma, #M7167-100ML) via gentle pipetting. The denuded oocytes were then collected, washed with PBS, and fixed in cold methanol (Sigma, #179337-500ML) for 30 min at room temperature.

Skeletal myofiber isolation

(i) Male C57BL/6J mice (6-8 weeks old) were used for muscle isolation. Mice were euthanized by cervical dislocation in accordance with institutional animal care and use guidelines prior to sample collection. Following established protocols (98), the extensor digitorum longus (EDL) and soleus muscles were carefully dissected from the hindlimb. Briefly, the dermal layer was excised, and the tibialis anterior muscle was delicately detached using fine forceps and surgically excised adjacent to the knee joint. The gastrocnemius muscle was carefully peeled back to expose the soleus muscle, which was subsequently isolated through precise tendon transection and meticulous separation from the underlying osseous structure. (ii) Harvested muscles were immediately transferred to a 15 mL centrifuge tube containing a specialized digestion solution (comprising 2% Collagenase Type I (Gibco, #17100017), 4.4 mL DMEM (Gibco, #11965092), and 0.1 mL 100 mM sodium pyruvate (Gibco, #11360070)). Enzymatic digestion was conducted at 37°C for 30-60 min, with intermittent tube inversion at 10-min intervals. Microscopic monitoring ensured complete

dissociation of skeletal muscle fibers. (iii) Enzymatically processed muscle fibers were transferred to a pre-chilled Petri dish containing a refined wash buffer (comprising 4.45 mL DMEM (Gibco, #11965092), 0.5 mL fetal bovine serum (HyClone, HYCLSH, #30088.02), and 0.05 mL 100 mM sodium pyruvate (Gibco, #11360070)). Employing gentle trituration via a Pasteur pipette, individual muscle fibers were systematically liberated. Morphologically intact fibers were selectively harvested and transferred to a fresh wash buffer dish, undergoing two consecutive washing cycles. A terminal wash with pre-cooled PBS was executed, maintaining approximately 1 mL of residual buffer. (iv) Muscle fibers were transferred to a methanol-treated dish. Under gentle agitation, 1 mL of pre-cooled methanol was added, creating a 50% methanol mixture and fixed at -20°C for 3 min. The methanol concentration was sequentially increased by removing PBS-methanol mixture and adding pure methanol, with fixation repeated at 75% and subsequently at near-100% methanol concentrations. After fixation, fibers were allowed to settle, and excess methanol was carefully removed, leaving sufficient volume for subsequent chip application. Using a wide-bore pipette tip, 30-50 µL of fixed muscle fibers were carefully spread onto the designated chip surface.

Flow cytometric analysis of cell proportion in PBMCs

The PBMCs were incubated with antibodies against CD14-APC (BD, #557831), CD3-APC (BD, #565119), CD16-APC (BioLegend, #302012), and CD19-FITC (Invitrogen, #11-0199-42) for 30 min at 4°C in darkness. Following incubation, the PBMCs were washed with PBS and stained with 1 µg/µL DAPI (BD, #564907) to identify dead cells. The cells were subsequently processed using a FACS Aria III (BD Biosciences). Single cells were identified based on forward scatter area versus height parameters, and all subsequent analyses were conducted on the total single-cell population without additional gating. Flow cytometric data were analyzed using FlowJo software version 10.1 (BD).

The smRNA-FISH assay of oocytes

The smRNA-FISH assay was performed as previously described, with some modifications (97). Oocytes were fixed with 4% paraformaldehyde for 20 min at room temperature, followed by extensive washing with homemade wash buffer containing 0.1% Triton X-100 (Sigma, #T8787) in PBS. Permeabilization was done with 1% Triton X-100 for 30 min at room temperature followed by another wash with the homemade buffer. Hybridization was carried out using the Fluorescent In Situ Hybridization Kit (Ribo-bio, #C10910) and FISH Probe Mix (Ribo-bio, #Inc1032792 for *Slc45a3* and #Inc1032793 for *Ooep*) according to the manufacturer's instructions. Both the pre-hybridization and hybridization solutions were supplemented with 0.1% Triton X-100. Pre-hybridization was performed at 37°C for 30 min, followed by hybridization at 37°C for 3 hours. Oocytes were subsequently washed at 42°C three times with 4x SSC containing 0.1% Tween-20, then once each with 2x SSC and 1x SSC, both supplemented with 0.5% BSA, and finally with the homemade wash buffer, each wash lasting 5 min. Finally, the oocytes were stained with DAPI (BD, #564907) in homemade wash buffer for 20 min at room temperature to label DNA, followed by three washes with the homemade wash buffer, each for 5 min.

The RNAscope assay of myofibers

Gradient fixation: Freshly extracted skeletal myofibers were resuspended in approximately 500 µL of PBS within a 6-cm dish. Pre-cooled methanol was incrementally introduced, with 500 µL of solution systematically discarded after thorough mixing. The fixation protocol involved sequential incubation at -20°C for 3 min, progressively increasing methanol concentrations to approximately 50, 70, 85, and 100%.

RNAscope: RNAscope immunofluorescence was executed utilizing the Multiplex Fluorescent Reagent Kit v2 [Advanced Cell Diagnostics (ACD), Bio-Techne, #323100] and Motic Custom PA53 FS6 microscopy,

according to the manufacturer's instructions for fresh frozen sample sections with several modifications. The staining procedure commenced with Protease Plus (ACD) treatment for 10 min at 40°C, following preliminary hydrogen peroxide (ACD) preprocessing, as referenced in the methodology by De *et al.* (99). For probe hybridization, a carefully calibrated mixture of RNAscope probes Mm-Myh4-C1 (ACD, #452731) and Mm-Myh1-C3 (ACD, #828491-C3) was applied at a 50:1 dilution ratio. Tyramide Signal Amplification (TSA) Vivid dyes were employed for horseradish peroxidase (HRP) probe conjugation. Specifically, TSA Vivid Fluorophore 570 (ACD, Bio-Techne, #PG-323272) was diluted 1:2500 in TSA buffer and conjugated with C1, while TSA Vivid Fluorophore 520 (ACD, Bio-Techne, #PG-323271) was diluted 1:1500 and conjugated with C3.

Stereo-cell chip preparation

PLL pre-coating was performed on Stereo-seq chips (STOmics, #200CT114) as follows: (i) The chip was placed in a 24-well plate (Thermo, #142485) (the following reagent volumes are based on the S1 chip as an example). The chip surface was washed twice with 1 mL of nuclease-free water (Invitrogen, #AM9932), then the back and edges of the chip were gently wiped using dust-free paper (Kimberly-Clark, #0130-10). To ensure complete drying, the chip surface was blown with compressed air until no liquid residue remained. (ii) After transferring the chip to a new well, 400 μ L of 0.01% poly-L-lysine solution (Sigma, #25988-63-0) was carefully added to the chip surface. Following a quick manual shake, the chip was placed on a shaker (IKA, #0003617000) and oscillated at 50-250 rpm at 23°C for 30 min. (iii) The poly-L-lysine solution was removed, the chip was washed gently as in step i, and then transferred to a clean well.

Stereo-cell library preparation and sequencing

Cell dropping, capture and fixing: (i) 20 μ L of cell suspension was carefully dropped onto the surface of the PLL-precoated chip and gently spread with a pipette tip to ensure even distribution, avoiding direct contact with the chip surface. Cells were incubated on the chip for 10 min to allow adhesion and capture. (ii) After incubation, the chip was placed into pre-cooled methanol at -20°C for 30 min to fix the cells. (iii) Excess methanol was carefully removed, and the chip was dried before proceeding to the next steps.

Cell staining and imaging: (i) For DiI staining, 5 μ M DiI (MedChemExpress, #41085-99-8) was prepared in 1 mL methanol and pre-cooled to -20°C. DiI staining was performed concurrently during the fixation process. (ii) For mIF staining, all steps were carried out at room temperature. After fixation, blocking was performed by adding a blocking buffer containing 5x SSC Buffer (Thermo, #AM9770), Hi-FBS, and nuclease-free water to the chip, followed by a 15-min incubation. After removing the blocking buffer, primary antibodies—WGA-Alexa FluorTM 555 conjugate (Thermo, #W32464), Anti-CD3 (Abcam, #ab245731), and Anti-CD45RA (BioLegend, #304102)—were added at appropriate concentrations in blocking buffer and incubated for 45 min. The chip was then washed twice with 0.1x SSC Buffer for 3 min each. Secondary antibodies—goat anti-rabbit Alexa Fluor Plus 647 (Thermo, #A48285) and goat anti-mouse Alexa FluorTM Plus 488 (Thermo, #A32723TR)—were diluted in 3x SSC Buffer and added to the chip, followed by a 30-min incubation. The chip was then washed as described above. (iii) For trypan blue staining, a 1:1 mixture of trypan blue solution (Sigma, #T8154-20ML) and 5x SSC Buffer was used. For DAPI staining, 5x SSC Buffer containing 1 μ g/ μ L DAPI (BD, #564907) was used. Both staining solutions were supplemented with 0.05 U/ μ L RNase inhibitor (RI) to a total volume of 200 μ L. The staining solution was applied to the chip surface and incubated in the dark for 5 min at room temperature. After staining, the chip was washed twice with 0.1x SSC Buffer supplemented with 0.05 U/ μ L RI. (iv) For imaging, 5 μ L of glycerol (Sangon Biotech, #4100854050) was applied to the chip surface,

and a coverslip was gently placed to ensure full glycerol infiltration. The chip was incubated at room temperature for 5 min before imaging with a Motic Custom PA53 FS6 microscope. After imaging, the chip was incubated in 5x SSC Buffer with 0.05 U/ μ L RI for 5 min and then washed once with 0.1x SSC Buffer with 0.05 U/ μ L RI.

Cell permeabilization and probe hybridization: (i) Cells were permeabilized by incubation with 0.1% pepsin (Sigma, #P7000) for 30 s and then washed with 0.1x SSC Buffer supplemented with 0.05 U/ μ L RI. (ii) A pre-hybridization solution (200 μ L) containing 20% formamide (VWR, #0606), 2x SSC Buffer, and 0.05 U/ μ L RI was slowly and gently added to the chip surface and incubated at 55°C for 5 min. After removing the solution, 200 μ L of 0.1x SSC Buffer supplemented with 0.05 U/ μ L RI was added and incubated at 4°C for 5 min, followed by a final wash with the same buffer.

In situ reverse transcription, amplification of released cDNA, and sequencing

Reverse transcription was performed on-chip at 42°C for 3 hours using 200 μ L Golden Reverse Transcriptase Mix (GCATBio, #LS-EZ-E-000240), containing 10 U/mL Golden Reverse Transcriptase, 1 mM dNTPs, 1 M betaine, 7.5 mM MgCl₂, 5 mM DTT, 2 U/mL RI, and 1x First-Strand buffer, supplemented with 2.5 mM TSO (5'-CTGCTGACGTACTGAGAGGC/rG/rG//iXNA_G/-3'). After reverse transcription, cDNA was released using a cDNA Release Mix (STOmics, #211KN114-EA) composed of cDNA Release Enzyme (STOmics, #1000028511) and buffer (STOmics, #1000028512), incubated at 55°C for 3-18 hours. Released cDNA was purified using 0.7x VAHTS DNA Clean Beads (Vazyme, #N411-03). Purified cDNA was resuspended in 200 μ L of amplification mix containing 1x Platinum High-Fidelity ReadyMix (GCATBio, #LS-EZ-K-000080) and 0.8 μ M cDNA PCR primer (5'-CTGCTGACGTACTGAGAGGC-3'). PCR cycling conditions were: 95°C for 5 min; 12 cycles (16 cycles for PBMC samples) of 98°C for 20 s, 58°C for 20 s, 72°C for 3 min; and a final extension at 72°C for 5 min. Amplified cDNA was purified with 0.7x DNA Clean Beads. Sequencing libraries were quantified using the Qubit ssDNA Assay Kit (Thermo, #Q10212). 80 ng of cDNA from each sample was used to generate DNA nanoballs (DNBs), which were sequenced on the MGI DNBSEQ-Tx platform with 50 bp for read 1 and 100 bp for read 2.

Cell cultivation on Stereo-cell chip

(i) For in situ sequencing of cultured cells, 1,000 NIH-3T3 cells were directly cultured on the surface of a sterile S1 chip. After cell dropping, smearing, and a 10-min incubation on the chip surface, the chip was transferred into a sterile 24-well plate (Thermo, #142485). One milliliter of complete growth medium—consisting of 90% DMEM, 10% Hi-FBS, and 1x penicillin-streptomycin—was added, and cells were cultured for 64 hours. (ii) For lipopolysaccharide (LPS) stimulation of NIH-3T3 cells during cultivation, LPS solution (Solarbio, #L4391-1MG) was added to the 1 mL complete growth medium 4 hours prior to the end of the 64-hour incubation, to a final concentration of 100 ng/mL. Cells were fixed immediately after cultivation. (iii) To evaluate the efficiency of nucleic acid capture under culture conditions, 1,000 NIH-3T3 cells were first cultured on the chip for 64 hours, followed by in situ membrane staining using DiI (MedChemExpress, #41085-99-8) and imaging with a Motic Custom PA53 FS6 microscope. Subsequently, cells were digested and removed from the chip using 25% Trypsin-EDTA (Biosharp, #BL512A). Next, 20 μ L of 5x SSC buffer containing oligonucleotides with 18 specific identifier barcodes and poly(A) tails, diluted across a concentration gradient, was dropped onto the surface of the PLL-precoated S1 chip. Stereo-cell was then used for in situ oligo capture and sequencing. To estimate cell coverage areas, the distribution of DAPI-stained nuclear signals was used. Region boundaries were expanded outward by 10, 20, 30, and 40 DNB distances to define four progressively distant uncovered regions at 10-DNB intervals (approximately 5 μ m each).

EVs staining and imaging on Stereo-cell chip

(i) For in situ staining of EVs on Stereo-cell chip, 1,000 NIH-3T3 cells were cultured on the surface of a sterile chip for 64 hours. (ii) Following two washes with 1 mL of PBS, cells were incubated with 500 μ L pre-chilled blocking buffer (10% FBS in PBS) for 15 min on the chip. The chip was subsequently transferred to a fresh well, and primary antibodies were introduced: Calnexin Monoclonal antibodies (Proteintech, #66903-1-IG-100 μ L) and SYN1-Specific Rabbit PolyAb (Proteintech, #20258-1-AP), diluted to one percent concentration in blocking solution, and incubated for 1 hour to facilitate specific molecular targeting. (iii) Following primary antibody incubation, the chip underwent two PBS washing cycles and was subsequently fixed with methanol at -20°C for 30 min. (iv) Secondary antibody conjugation was performed using Goat Anti-Rabbit IgG H&L (Alexa Fluor® 488) (Abcam, #ab150077) and DyLight 647 Conjugated AffiniPure Goat Anti-Mouse IgG (H+L) (BosterBio, #BA1151). These secondary antibodies were added at one percent concentration in 3x SSC buffer and incubated for 40 min, followed by two washing steps with 0.1x SSC buffer to remove non-specifically bound antibodies. (v) The chip was then incubated with 1 μ g/ μ L DAPI in 3x SSC for 5 min, washed twice with 0.1x SSC, and imaged with the Motic Custom PA53 FS6 microscopy.

Stereo-cell-CITE

(i) PBMCs were resuspended in complete growth medium at a concentration of 1 million/mL and divided into two groups. The experimental group was treated with 1x Cell Stimulation Cocktail (TONBO Biosciences, #TNB-4975), while the control group received an equivalent volume of DMSO (Sigma, #D2650). After 6 hours of incubation in a cell culture incubator, the cells were washed twice with PBS. (ii) The PBMCs were resuspended in 45 μ L of Cell Staining Buffer (BioLegend, #420201) and then incubated with FcR block reagent (BioLegend, #422301) at 4°C for 10 min. The TotalSeq-A antibody pool (BioLegend, #399907) was prepared according to the manufacturer's instructions. The blocked cell suspension was then mixed with 25 μ L of TotalSeq-A antibodies and incubated on ice for 1 hour. Subsequently, the cell suspension was washed three times with 1 mL of PBS and centrifuged at 400 g for 5 min at 4°C. (iii) Cell viability was assessed using trypan blue staining (Sigma, #T8154-20ML), and the concentration was adjusted to 500 cells/ μ L with PBS (Meilunbio, #MA0015). A 20 μ L aliquot of the diluted cell suspension was applied to the PLL pre-coating chip. The subsequent experimental steps followed the Stereo-cell library preparation until the cDNA was released from the chips. (iv) The cDNA and ADT were purified using 1x VAHTS DNA Clean Beads (Vazyme, #N411-03). cDNA libraries were constructed as previously described, while ADT products were amplified using the kit (MGI, #202KA114) with the following cycling conditions: 95°C for 5 min, 23 cycles of 98°C for 20 s, 58°C for 20 s, 72°C for 30 s, and a final extension at 72°C for 5 min. The amplified ADT was purified with 2x VAHTS DNA Clean Beads. Eighty nanograms each of cDNA and ADT were used for generating DNBs, with rolling circle amplification times of 30 and 35 min, respectively. DNBs were quantified using the Qubit ssDNA Assay Kit (cat# Q10212, Thermo). (v) cDNA and ADT libraries were sequenced by the MGI DNBSEQ-Tx sequencers with paired-end reads. cDNA libraries were sequenced with a read length of 50 bp for read 1, 50 bp for read 2, and 10 bp for the sample barcode, while ADT libraries were sequenced with read length 50 bp for read 1, and 36 bp for read 2.

Raw sequencing data preprocessing

To process the raw data from Stereo-cell, we aligned the Coordinate ID (CID) sequences from read 1 to the pre-determined list of barcodes with known coordinates, permitting a single base mismatch. Only reads with a valid CID and a high-quality UMI (which includes a maximum of two bases with a Phred score no higher than 10, and no N bases) were converted into a FASTQ+ format. In this format, the CID and UMI were appended to the sequence identifiers (100). The

formatted single-end FASTQ+ file was then mapped to the mm10 genome using STAR, but only specifically mapped reads were kept for the purpose of gene annotation and read counting (101, 102). Reads with a mapping score below 10 were discarded. In the final step, we counted the UMI-deduplicated read numbers for each gene and each CID, and compiled this data into a Matrix Market (MM) file with five columns: gene ID, x coordinate, y coordinate, UMI count, and UMI count within introns.

Staining image registration and quantification

To register the staining image with the UMI, TrakEM2 within ImageJ (v1.53f51) software was utilized. Initially, the spatially resolved gene expression matrix was transformed into a grayscale image format. This was then loaded into a fresh TrakEM2 project, along with the staining image or images. Once the images were unlocked, we designated the UMI image as the red channel and the staining image as the green channel. During the transformation process, we adjusted the position and size of the green channel so that it matched the red channel accurately. Finally, the aligned green channel, which represents the staining image, was saved as a newly registered flat image (103, 104).

Cell segmentation

UMI-based cell segmentation: A grayscale image was created from the MM file, with each pixel representing a CID. The brightness of each pixel corresponded to the UMI count for all genes associated with that CID. ImageJ software was used to automatically adjust the image's contrast and brightness by analyzing the image histogram. Next, we utilized the UNet++ framework to conduct cell segmentation (20). This was followed by a watershed analysis to separate clustered cells in the segmentation results (105). Image dilation was applied to improve the contours of the segmentation. Finally, a new column was appended to the MM file to denote the cell labels. The code used for UMI-based cell segmentation is indexed in Zenodo (106)

DAPI-based cell segmentation: To separate the segments containing multiple cells, we utilized the advanced deep learning toolkit StarDist. This toolkit comes with a built-in model named “2D_versatile_fluo,” which is specifically pre-trained to handle the segmentation of nuclear fluorescence staining and was applied to our DAPI-stained images (107). Given the extensive size of the images, we made use of the predict_instance_big function, which is tailored for processing large-scale images. We also adjusted the n_tile parameter to (8, 8), meaning the image was divided into 64 (8 x 8) sections for processing each section one at a time. Subsequently, we performed an image dilation process on the muscle nuclei segmentation to enlarge the nuclear areas.

Muscle ROI parcellation: To segment muscle fibers into spatially uniform ROIs, we initiated the process by manually delineating each individual muscle fiber boundaries. Utilizing the K-means clustering algorithm, we systematically grouped DNBs within each fiber, employing their spatial coordinates as algorithmic input. The number of clusters was algorithmically determined by dividing the total muscle fiber area by 6,400, resulting in ROIs with a standardized average area of 6,400, corresponding to an 80 x 80 bin spatial resolution.

Oocyte cell segmentation: We produced a grayscale image for the purpose of oocyte segmentation by manually utilizing the MM file. This was followed by an automated enhancement of the image using ImageJ. With the help of ImageJ's oval selection instrument, we individually marked every oocyte present in the image. Subsequently, we exported the coordinates that encapsulated each oocyte and provided a distinct identifier to every one of them. These identifiers were also incorporated into the MM file.

Segmentation accuracy assessment

Each DNB within a range of 100 DNB units from the center of every segmentation was evaluated and categorized into 100 equal-sized groups. The UMIs for each group was then computed as a percentage of the total UMI obtained by the entire chip. In a similar manner, the percentage of the chip's total capture area pertaining to each group was also calculated. Following this, the accumulated proportions of UMI capture and area coverage for these groups were graphically represented on a line chart. The blue dot located above the chart's line is indicative of a segmentation being of good quality.

Single-cell clustering

The MM file facilitated the generation of a cell-by-gene matrix which was then processed using Seurat for advanced examination. Prior to employing the Seurat protocol, scDBlFinder (21) was utilized to eliminate potential doublets, and any segmentation spanning multiple DAPI-stained cells was also discarded. The Seurat workflow comprised six primary operations: (1) Data normalization via the NormalizeData function, (2) Selection of variable features using the FindVariableFeatures function, (3) Data scaling with the ScaleData function, (4) Dimensionality reduction through the RunPCA function, (5) Batch effect mitigation, (6) Detection of cell neighbors with the FindNeighbors function, (7) Community clustering employing the FindClusters function, and (8) Two-dimensional cell projection using UMAP (RunUMAP function) (108). Except for the fifth step, all procedures were uniformly applied to samples of PBMC, NIH-3T3, muscle fibers, and oocytes. For the PBMC samples for multiple platform comparison, canonical correlation analysis (CCA) was utilized for batch effect correction using the FindIntegrationAnchors and IntegrateData functions. Publicly available 10x Genomics PBMC data (10k PBMCs, v3 chemistry) can be accessed at: https://support.10xgenomics.com/single-cell-gene-expression/datasets/3.0.0/pbmc_10k_v3. NIH-3T3 and oocyte samples underwent harmony analysis for batch effect resolution (109). Additionally, the diffusion map dimension reduction analysis was executed to delineate the oocyte state transition process (110).

Stereo-cell-CITE data analysis

We independently implemented the Seurat computational workflow. For protein data, we utilized Harmony to correct batch effects, while for RNA data, we employed CCA within the Seurat framework to mitigate batch effect. We subsequently performed a weighted nearest neighbor (WNN) analysis, integrating RNA and protein datasets by incorporating the first 20 Harmony dimensions of protein data and the first 30 Principal Component Analysis (PCA) dimensions of RNA data. Following integration, we re-embedded cells into a new UMAP dimensional space using a weighted combinatorial approach of RNA and protein datasets. Subsequently, we leveraged CiteFuse (111) to infer the protein-RNA interaction network, first converting the Seurat object to a SingleCellExperiment (112) object as a prerequisite. Finally, we employed the igraph (113) package to visualize and manipulate the inferred molecular interaction networks.

EV inference

Utilizing the analytical capabilities of SEVtrass (56), we systematically identified putative EVs within the on-chip cultured NIH-3T3 Stereo-cell dataset. Initially, for each individual experiment, we transformed our data into Matrix Market files. These files were then carefully stored in a specifically designated directory, which would later serve as the input for SEVtrass. Subsequently, we executed the seV_recognizer function to infer the presence of EVs. After successfully inferring the EVs, we visualized them using the UMAP plot generated from the Seurat workflow, enabling us to gain a more intuitive understanding of their distribution within the dataset.

Cell adjacency quantification

In R, we used the nn2 function from the RANN package, which utilizes a K-D tree structure (available at <https://github.com/jefferislab/RANN>),

to identify the ten closest neighboring cells for each cell. Following this, we grouped neighboring cells belonging to the same cluster according to their respective cluster. To assess the statistical significance of the tendency of cells from one cluster to be found near cells of another cluster, as well as to measure the fold change between experimental and random data, we conducted a binomial test using the binom.test function in R.

Spatially coexpressed gene module identification

We transferred the gene expression data and the spatial positions of each nucleus in muscle fiber datasets, or each bin 10 in oocyte datasets, into separate tables. These tables were then transformed into DataFrames in python. Subsequently, we constructed a hotspot object from these DataFrames (74). We then established a K Nearest Neighbors network, computed the pair-wise gene correlations, and performed gene clustering. For each nucleus or bin 10, we quantified the collective expression levels of the gene clusters, referred to as a module, using the calculate_module_scores function.

Trajectory inference

We utilized Monocle 3 to pinpoint the state transition of oocyte cells, which is referred to as the trajectory. Each cell was arranged on this trajectory and given a transition score, known as pseudotime (114). Through linear fitting, we detected genes that exhibited variations in expression levels corresponding with the trajectory. The pseudotime and trajectory sequence for each cell were incorporated into the Seurat object. Subsequently, the cells were segmented into 20 equal cohorts according to their trajectory sequence, and the mean expression levels of genes associated with the trajectory were computed for the purpose of visualization.

Regulatory network analysis

The analysis of the gene regulation network was conducted through the SCENIC framework (88). The process involved three stages: first, identifying co-expression modules with the help of GRNBoost2; secondly, analyzing the enrichment of motifs and their target genes within each module using Prune2df; and thirdly, computing the area under the curve (AUC) score for every cell in each module by employing Aucell. To visualize the results, cells were categorized according to their trajectory order, and their AUC scores were averaged.

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applications. Employees of BGI have stock holdings in BGI. **Data and materials availability:** Stereo-cell datasets have been deposited to the CNGB Nucleotide Sequence Archive (https://db.cngb.org/data_resources/project/CNP0005427/). Code for processing spatial sequencing libraries is available at Zenodo ([115](https://doi.org/10.1101/2024.06.12.603478)). All data are available in the main text or the supplementary materials. **License information:** Copyright © 2025 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/about/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

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Figs. S1 to S12; Tables S1 to S9

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